

2026**Computer Science and Application****Full Marks: 50****Pass Marks: 17****Time: 2 hours***(The figures in the margin indicate full marks for the questions)***Allotment of Marks**

Q. No. 1 carries 1 mark each	1 × 4 = 4
Q. No. 2 carries 2 marks each	2 × 8 = 16
Q. No. 3 carries 5 marks each	5 × 6 = 30
Total	Total = 50

1. Answer the following questions: 1 × 4 = 4

- (a) What is a database?
- (b) What is a field?
- (c) State the use of *break* statement in a *switch-case* statement.
- (d) What is a character constant in C++?

Turn Over

2. Answer the following questions: 2 × 8 = 16

- (a)** Write the differences between DML and DDL. **(2)**
- (b)** Write the use of the %= operator with an example. **(2)**
- (c)** Rewrite the following statement using an *if-else* statement:
`x = ((y < z) ? y : z);` **(2)**
- (d)** What is meant by testing and debugging in a program? **(2)**
- (e)** Differentiate between *while* and *do-while* loop. **(2)**
- (f)** What is a flowchart? Give one example. **(2)**
- (g)** Differentiate between `i++` and `++i`. **(2)**
- (h)** What is operator in C++? Explain briefly. **(2)**

3. Answer any six questions of the following: $5 \times 6 = 30$

- (a) Define data type. What are the different data types used in C++? **$1 + 4 = 5$**
- (b) What is constructor? Explain with an example. **$2 + 3 = 5$**
- (c) Explain the different types of errors in programming. **5**
- (d) Write a program in C++ to display all prime numbers within a given range. **5**
- (e) What is a variable? Why is a variable called symbolic variable? **$1 + 2 + 2 = 5$**
What is meant by dynamic initialization of a variable? Give example.
- (f) What is the role of comments and indentation in a program? **5**
- (g) What are data types in MySQL? Name some common data types and their uses. **$1 + 4 = 5$**
- (h) What is Typecasting? Mention the use of *sizeof* operator. **$2 + 3 = 5$**